FCFS Appointment Simulation: A Plausible Two-Class Parameterization

Supplementary comparison with the stress-test baseline

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Abstract

This is a supplementary note to the stress-test analysis. The stress-test baseline ($\lambda = 100$ arrivals per day, $S = 32$ slots, demand-to-capacity ratio 3:1) was designed to expose how simulation metrics behave under severe capacity pressure, not to represent a plausible clinical setting. This note re-runs the core diagnostics under a more plausible, though still stylized, two-class parameterization drawn from the model formulation note (v2): $\lambda = 24$ arrivals per day against $S = 20$ slots (1.2:1 ratio), a 28-day booking horizon, and behaviorally asymmetric classes representing an MRI-like diagnostic group and a behavioral-health follow-up group. Thirty independent seeds are used throughout.

Reducing demand pressure and introducing class asymmetry raises aggregate utilization to 0.934 and overall served rate to 0.789, substantially above the stress-test values. The more consequential change is a 22-percentage-point gap in class-level served rates (88.1% vs 66.1%) that is absent from the symmetric stress-test baseline. This gap is mechanically consistent with asymmetric behavioral parameters under a pooled FCFS booking pool: Class 1 has a higher balk tolerance, lower baseline no-show risk, and lower cancellation hazard than Class 2, and each advantage compounds under shared-capacity FCFS. The qualitative findings of the stress-test note survive: no-offer remains zero, offered wait exceeds accepted wait through the same selection argument, and no-show remains the direct slot-fill channel.

1 Parameter Configuration

Table 1 compares the two configurations. The most consequential differences are the demand-to-capacity ratio, the class asymmetry, and the behavioral thresholds. Under this parameterization, Class 1 (MRI-like) has longer balking and no-show thresholds (21 days vs 9 days and 6 days in the stress test), consistent with higher patient tolerance for long-wait diagnostic scheduling. Class 2 (behavioral health follow-up) is less tolerant (threshold at 14 days) and carries a higher baseline no-show probability below the threshold ($\xi_2^{\text{low}} = 0.15$, vs 0 in the stress test). Cancellation hazards are substantially lower (0.01–0.02/day vs 0.10/day). These parameters are taken from the model formulation note v2 and are illustrative, not empirically calibrated.

2 Outcome Comparison

Table 2 reports aggregate and class-level outcomes for both configurations, each averaged over 30 independent seeds.

Parameter	Stress test	Comparison
Total arrivals λ (per day)	100	24
Class 1 λ_1	50	14
Class 2 λ_2	50	10
Slots per day S	32	20
Demand/supply ratio	3.1	1.2
Booking horizon H (days)	14	28
Burn-in / cooldown (days)	30 / 10	60 / 30
Class 1 (MRI-like diagnostic)		
Balk threshold (days)	9	21
Low-delay balking prob. b_1^{low}	0.00	0.02
High-delay balking prob. b_1^{high}	0.50	0.55
No-show threshold (days)	6	21
Low-delay no-show ξ_1^{low}	0.00	0.01
High-delay no-show ξ_1^{high}	0.30	0.31
Daily cancel prob. ϕ_1	0.10	0.01
Class 2 (Behavioral health follow-up)		
Balk threshold (days)	9	14
Low-delay balking prob. b_2^{low}	0.00	0.04
High-delay balking prob. b_2^{high}	0.50	0.72
No-show threshold (days)	6	14
Low-delay no-show ξ_2^{low}	0.00	0.15
High-delay no-show ξ_2^{high}	0.30	0.51
Daily cancel prob. ϕ_2	0.10	0.02

Table 1: Configuration comparison. Stress-test parameters are from `configs/baseline.yaml`; comparison parameters from `configs/realistic.yaml`, following the illustrative two-class profile in the model formulation note v2. Both configurations are stylized.

3 What Changes and What Does Not

Aggregate utilization and served rate. Both improve substantially relative to the stress test. Utilization rises from 0.842 to 0.934; served rate from 0.269 to 0.789. The primary driver of the utilization gain is the much lower cancellation hazard: with $\phi = 0.01$–0.02/day instead of 0.10/day, far fewer booked slots are vacated before service. The approximate survival probability to service is $0.99^{10.9} \approx 0.90$ for Class 1 and $0.98^{10.9} \approx 0.80$ for Class 2 at mean $\tau \approx 10.9$ days, against $0.9^{8.35} \approx 0.42$ in the stress test. The served-rate gain also reflects the lower demand pressure: the structural ceiling is $20/24 = 0.833$, and the simulation output (0.789) is mechanically consistent with this bound net of no-shows. In the stress test the ceiling is 0.32, which the output never approaches.

Balking collapses; the offered/accepted gap nearly vanishes. In the stress test, mean offered wait (9.30 days) sits above the common 9-day balk threshold, driving a 35.6% balked share and a 0.95-day offered/accepted gap through high-τ patient selection. Under this parameterization, mean offered wait (10.87 days) remains well below the Class 1 threshold (21 days) and modestly below the Class 2 threshold (14 days). Only 3.3% of arrivals balk, and the offered/accepted gap nearly closes ($\Delta = 0.02$ days). The selection artifact that makes accepted wait unreliable in the stress test is

Metric	Stress test	Comparison
Aggregate outcomes		
Average utilization	0.8418	0.9341
Overall served rate	0.2692	0.7890
Mean offered wait (days)	9.30	10.87
Mean accepted wait (days)	8.35	10.85
Offered minus accepted wait (days)	0.95	0.02
Arrival outcome shares		
Served	0.269	0.789
Balked	0.356	0.033
Canceled	0.323	0.123
No-show	0.052	0.056
No-offer	0.000	0.000
Unresolved	<0.001	0.000
Class-level served rates		
Class 1 served rate	0.269	0.881
Class 2 served rate	0.269	0.661
Class gap (C1 − C2)	0.001	0.220

Table 2: Aggregate and class-level outcomes under both configurations, averaged over 30 independent seeds. Stress-test figures are from the controlled 30-seed diagnostic run reported in the first note.

mechanically absent here.

No-show share is similar; the mechanism differs. No-show shares are similar (5.2% vs 5.6% of arrivals), but for structurally different reasons. In the stress test, mean accepted delay (8.35 days) exceeds the 6-day no-show threshold, so the fraction of booked patients with $\tau > 6$ is substantial and those patients face $\xi^{\text{high}} = 0.30$ no-show probability at service time. Under this parameterization, the 21-day Class 1 threshold means that the large majority of Class 1 bookings are placed at $\tau < 21$ days and face only $\xi_1^{\text{low}} = 0.01$. The aggregate no-show share is pulled up by Class 2, for which $\xi_2^{\text{low}} = 0.15$ applies even to short-wait bookings. This makes Class 2 a persistent source of wasted booked capacity across the delay distribution, not only at long delays.

A large class access gap emerges. The simulation output shows a 22-percentage-point gap in class-level served rates (88.1% vs 66.1%), which is absent from the symmetric stress-test baseline (gap = 0.001) and substantially larger than the gaps produced by the stress-test sensitivity screen under one-at-a-time parameter perturbations. This gap is mechanically consistent with three compounding behavioral asymmetries. Class 1 has a higher balk tolerance (threshold 21 vs 14 days), allowing it to claim later-horizon slots that Class 2 declines. Class 1 has a lower baseline no-show probability ($\xi_1^{\text{low}} = 0.01$ vs $\xi_2^{\text{low}} = 0.15$), so its bookings are more likely to reach service. Class 1 also has a lower daily cancellation rate ($\phi_1 = 0.01$ vs $\phi_2 = 0.02$), reducing cumulative cancellation exposure. Under pooled FCFS, none of these differences confer explicit priority, but all three translate into access advantages because they affect slot consumption and booking survival at each stage of the pipeline.

Figure 1 shows the class-level served rates alongside the stress-test baseline for reference. The aggregate values look healthy; the class gap is not visible from aggregate metrics alone.

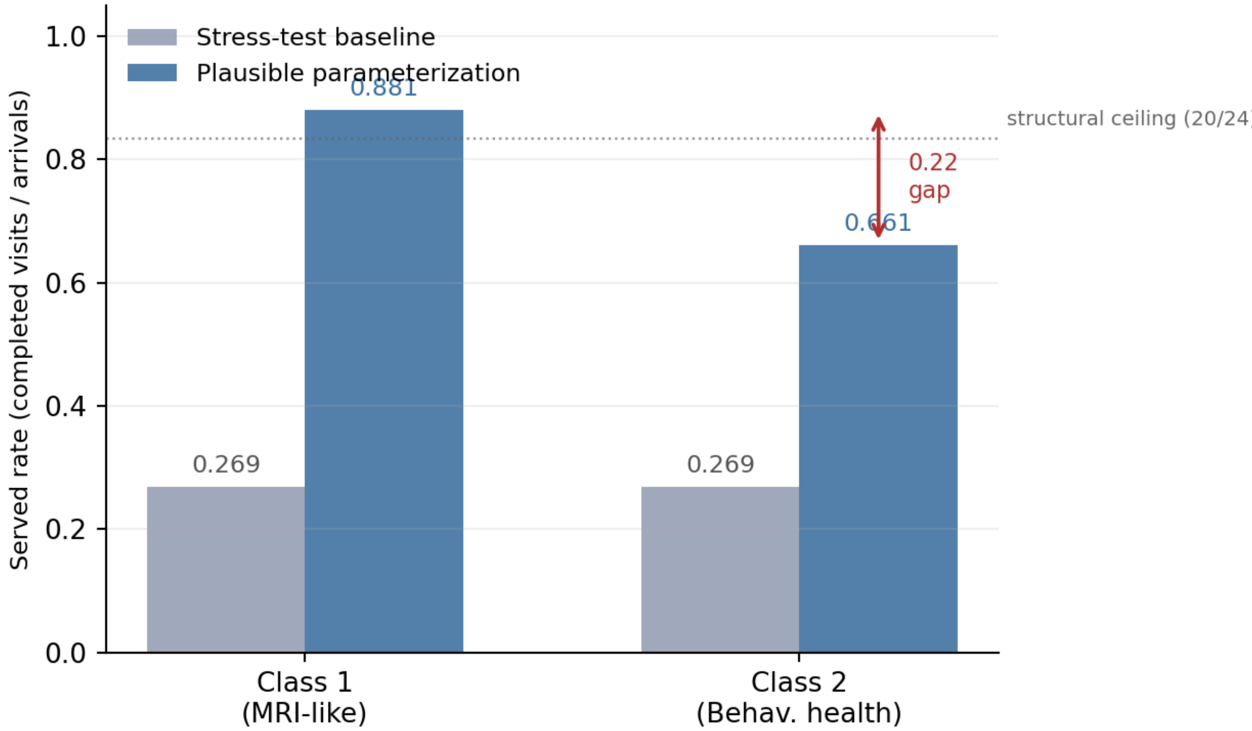

Figure 1: Class-level served rates under the plausible two-class parameterization (dark bars), compared with the stress-test baseline (light bars). Aggregate access improves relative to the stress test, but access is uneven across classes. Under pooled FCFS, this gap is mechanically consistent with asymmetric behavioral parameters rather than explicit class priority. The dotted line marks the structural served-rate ceiling for the comparison parameterization ($20/24 \approx 0.833$).

No-offer remains zero; horizon dynamics carry over. As in the stress test, cancellations continuously free future slots and the horizon does not saturate, so no-offer is zero. The structural argument is the same: daily arrivals (24) do not consistently outrun the combined effect of cancellations freeing future capacity and new-slot rollover at the horizon boundary.

4 Summary

The stress-test note showed that low served rate can be mechanically consistent with high utilization when demand strongly exceeds capacity. This comparison shows that reducing demand pressure raises aggregate served rate and changes the loss-channel mix; balking collapses, cancellation falls, and no-offer remains zero; but it does not eliminate class-level access differences.

Under the current pooled FCFS formulation, the simulation output is consistent with a 22-percentage-point class access gap (Class 1: 88.1%, Class 2: 66.1%) driven entirely by asymmetric behavioral parameters. Each asymmetry ; balk tolerance, baseline no-show risk, cancellation rate; operates through the shared booking pool and compounds without any explicit class prioritization. This gap is invisible in the aggregate utilization (0.934) and overall served rate (0.789), which look healthy. Aggregate metrics are not sufficient to diagnose access equity in this formulation.

The qualitative findings of the stress-test note carry over: offered wait remains the less biased delay measure, no-show is the direct slot-fill channel, and no-offer is zero under this parameterization. What this comparison adds is the demonstration that FCFS neutrality does not imply outcome equality when patient classes differ in behavioral risk, and that this effect can be quantitatively substantial even at moderate demand-to-capacity ratios.